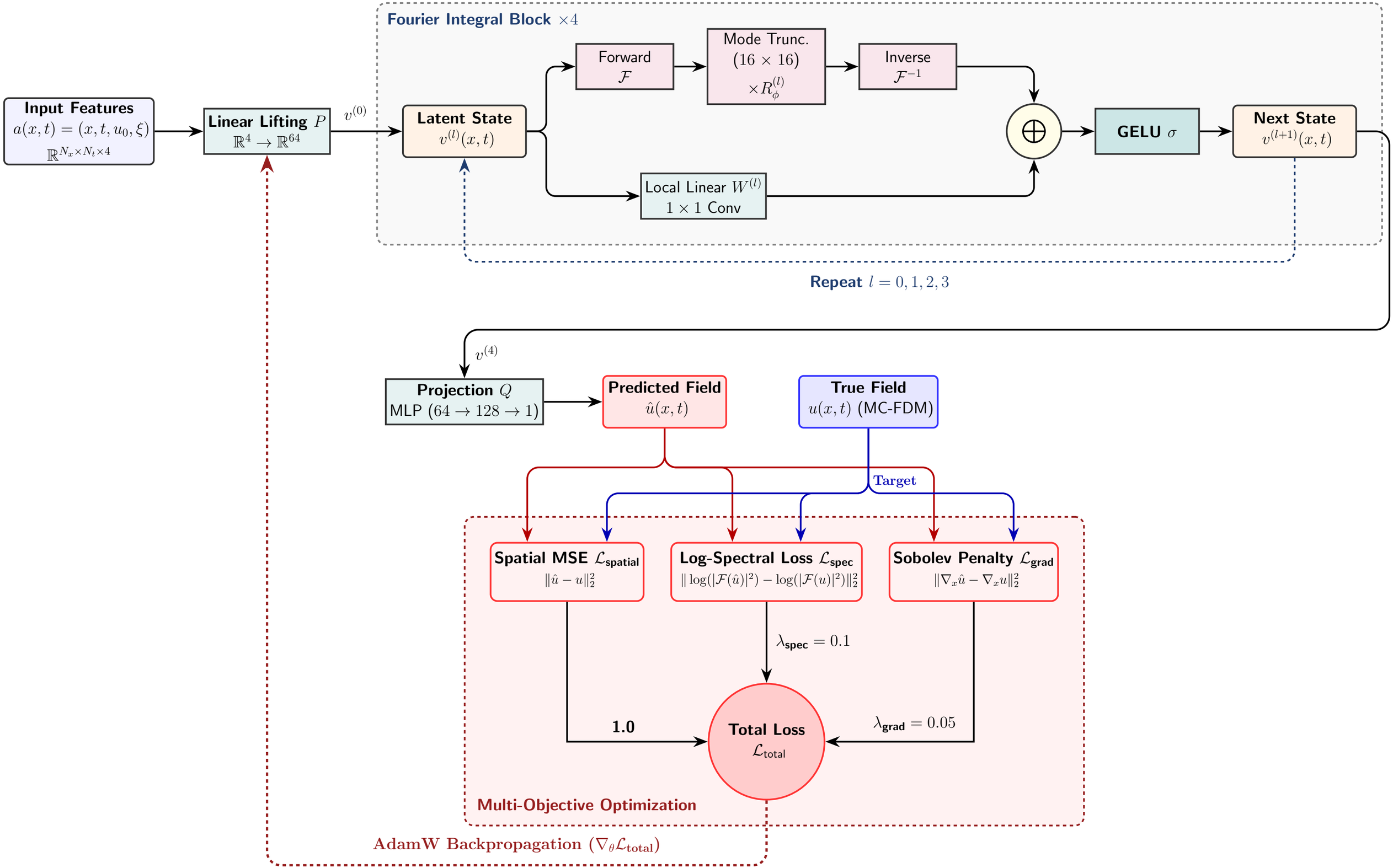

Fourier Integral Block $\times 4$
Input Features
$a(x,t) = (x,t,u_0,\xi)$
$\mathbb{R}^{N_x \times N_t \times 4}$
Linear Lifting P
$\mathbb{R}^4 \to \mathbb{R}^{64}$
$v^{(0)}$
Latent State
$v^{(l)}(x,t)$
Forward
$\mathcal{F}$
Mode Trunc.
(16×16)
$\times R_\phi^{(l)}$
Inverse
$\mathcal{F}^{-1}$
Local Linear $W^{(l)}$
1×1 Conv
GELU σ
Next State
$v^{(l+1)}(x,t)$
Repeat $l = 0,1,2,3$
$v^{(4)}$
Projection Q
MLP $(64 \to 128 \to 1)$
Predicted Field
$\hat{u}(x,t)$
True Field
$u(x,t)$ (MC-FDM)
Target
Spatial MSE $\mathcal{L}_{\text{spatial}}$
$\|\hat{u} - u\|_2^2$
Log-Spectral Loss $\mathcal{L}_{\text{spec}}$
$\| \log(|\mathcal{F}(\hat{u})|^2) - \log(|\mathcal{F}(u)|^2) \|_2^2$
Sobolev Penalty $\mathcal{L}_{\text{grad}}$
$\|\nabla_x \hat{u} - \nabla_x u\|_2^2$
$\lambda_{\text{spec}} = 0.1$
1.0
Total Loss
$\mathcal{L}_{\text{total}}$
$\lambda_{\text{grad}} = 0.05$
Multi-Objective Optimization
AdamW Backpropagation ($\nabla_\theta \mathcal{L}_{\text{total}}$)